Digital Image Processing (DIP)
Most Important 20 Questions (Unit 1 & Unit 2)

UNIT 1

1. Describe the Fundamental Steps in Digital Image Processing using a block diagram.
2. Elaborate on Image Sampling and Quantization.
3. Explain Image Sensing and Acquisition.
4. Discuss Basic Relationships between Pixels (Adjacency & Connectivity).
5. Explain the Components of a General-Purpose Image Processing System.
6. Define Digital Image Processing and distinguish it from Computer Vision.
7. Discuss the Elements of Visual Perception (rods and cones).
8. Explain Spatial Resolution and Intensity Resolution.
9. Calculate storage required for a digital image (e.g., 512×512, 256 gray levels).
10. Explain and compare Euclidean, City-block, and Chessboard distances.

UNIT 2

1. Explain Histogram Equalization with mathematical steps.
2. Explain Linear Filtering with a suitable diagram.
3. Compare Smoothing (Low-pass) and Sharpening (High-pass) filters.
4. Explain Correlation vs Convolution.
5. Explain Median Filter and its advantages over Mean filter.
6. Define Point Operations (gain, bias) and explain brightness enhancement.
7. Explain Gamma Correction and its effect on images.
8. Discuss the Sobel operator and gradient detection.
9. A noisy image contains impulse noise – justify using Median filtering.
10. A dark image needs enhancement – explain point processing method.